Dr. TIANYA HOU

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RESEARCH INTEREST

Data Mining, Machine Learning, Data Visualization, Economics, Social Data Science

EDUCATION

The Hong Kong Polytechnic University, Hong Kong SAR *Sep 2011 - Sep 2017*

- PhD in Department of Building and Real Estate
- Major in Applied Data Science in Urban Analytics, Social Data Science

Ajou University, Korea *Sep 2007 - Sep 2009*

- Master in Department of Industrial Engineering
- Major in Data Mining, Machine Learning, Economics

EXCHANGE PROGRAM

Vienna University of Economics and Business, Austria *Sep 2014 - Mar 2015*

- Eurasia-Pacific Uninet PhD Program
- In Department of Socioeconomics

RESEARCH EXPERIENCE

PhD Research, Mobility Analysis *Sep 2011 - Sep 2017*

The Hong Kong Polytechnic University, Hong Kong SAR

- A collaboration project between the university and the government, with the aim to make cities and housing market inclusive and sustainable.
- Identified mobility patterns of Hong Kong residents using "Link Analysis", and detected urban communities based on mobility behaviors using "Louvain Clustering".
- Most importantly, studied mobility from the gender perspective, offered a clearer understanding of women's needs, usage and preferences for urban mobility and how they differed from men.
- Visualized those patterns to understand the intersection between people and housing market in the effort to drive city more livable and sustainable.
- Achieved the Smart City Challenge Award 2015 finalist (100 of 3000) because of the importance and great contribution of the project.

Research Assistant, Housing Market Analysis *May 2010 - May 2011*

The Hong Kong Polytechnic University, Hong Kong SAR

- A collaboration project between the university and the company, to analyze the housing market in Hong Kong by using machine learning techniques.
- Predicted housing price based on the historical data using Sensitivity Analysis of Artificial Neural Network with the accuracy of 91.2%.
- Conducted housing demand analysis using Association Analysis to understand the expectations and perceptions of different household types.
- Enabled stakeholders to improve the investment decision process, optimize returns, and achieve a sustainable housing design based on users' preferences.

Master Research, Oil Price Prediction *Sep 2007 - Sep 2009*

Ajou University, Korea

- A collaboration project between the university and the government, to provide an accurate forecasting framework which can effectively capture the trend of the oil prices changes.
- Generated Technical Indicators from time series data (historical supply, demand, and economical data), to capture the time associations of the data at the same time remove the bad effects of the noises.
- Performed Nonlinear Principle Component Analysis to reduce the data dimensionalities thus enhanced the performance and robustness of the subsequent model.

- Predict oil prices using Graph based Semi-supervised Learning with the enhanced accuracy of 86% compared with other baseline algorithms.
- From the data science perspective, the great contribution of this research is to adjust Graph based Semi-supervised Learning algorithm to fit with time series input.

TEACHING EXPERIENCE

Teaching Assistant *Sep 2007 - Sep 2009*

Ajou University, Korea

- Courses included: Advanced Engineering Economy, Data Mining, Data Analysis & Practice, Statistics, and Operation Research.
- Responsibilities included: Ran the class in the event that the teacher is absent; Supervised undergraduate students working on research projects; Provided feedback on their research proposals and reports; Helped the teacher grade papers and tests.

Lab Instructor

Alpha Data Mining Lab, Ajou University, Korea

- Taught "Predictive Modeling with SAS Enterprise Miner";
- Provided hand-on experiences on SAS programming skills; Cultivated students' abilities to solve real world problems using data mining and machine learning techniques.

INDUSTRY EXPERIENCE

Data Scientist *Sep 2018 - now*

North China Electric Power Company, China

- Conducted data science projects with multidisciplinary teams, identified the expectations of the stockholders, as well as the limitations of the existing infrastructures.
- Monitored the performance of the systems, identified anomalies, and implemented necessary upgrades to optimize performance and reliability.
- Utilized advanced data science techniques to extract insights from the data, identified performance trends and opportunities for improvement, developed efficiency strategies.
- Attended training sessions to stay up-to-date on data science technologies and industry trends, participated in research to innovate and improve data analytics technologies.

HONOR & AWARD

Finalist of Hong Kong City Challenge (100/3000)	*2016*
Eurasia-Pacific PhD Scholarship	*2014*
Hong Kong RGC Studentship	*2011*
Brain Korea 21 Fellowship	*2007*
Kwang-Hua Scholarship	*2004*
Beijing Outstanding Graduate	*2003*

SERVICE & MEMBERSHIP

Decision Support System, Reviewer	*Since 2013*
The American Real Estate Society, Member	*Since 2014*
The Hong Kong PolyU CIB Student Chapter, Committee Member	*2012 - 2017*
Alpha Data Mining Lab (http://www.alphaminers.net), Member	*2007 - 2009*

SKILL AND CERTIFICATION

Programming Skills

- SAS
- C
- Matlab
- Python

Certification

- SAS BASE Certification
- National Rank Examination (C Language)

PUBLICATION

- Tianya Hou. Residential Mobility and Daily Mobility in Hong Kong. Ph.D Thesis, The Hong Kong Polytechnic University, 2017.
- Hyunjung Shin, Tianya Hou, Kanghee Park, Chan-Kyoo Park, and Sunghee Choi. Predication of movement direction in crude oil prices based on semi-supervised learning. Decision Support Systems, 55(1), pages 348-358, 2013. (co-first author)
- Tianya Hou. CRM using data mining techniques in real estate industry. The American Real Estate Society 30th Annual Meeting, Coronado (San Diego), CA, USA, 2014
- Tianya Hou and Hyunjung Shin. Influence of Noisy and High-Dimensional Data on Semi-Supervised Learning. The International Conference on Data Mining, USA, 2009.
- Tianya Hou, Kanghee Park, and Hyunjung Shin. Crude Oil Price Prediction Based on Semi-Supervised Learning. The 6th Business Intelligence World Conference, pages 418-436. 2009.
- Tianya Hou, Inae Choi, and Hyunjung Shin. How Robust Semi-Supervised Learning against Noise and High Dimensionality. The 5th Business Intelligence World Conference, pages 362-375. 2008.
- Tianya Hou. Equipment classification with unlabeled data and imperfect labels. Under revision.